

EcoPassEU: A Privacy-Aware Digital Product Passport DApp for Sustainable Consumer Goods in Europe

Group Number: 11

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1 What is the idea for your application?

EcoPassEU is a DApp designed to create and manage digital product passports (DPPs) for consumer goods, initially focusing on textiles and small electronics. By scanning a product-linked QR code or NFC tag, users can view a verified record of the item's provenance, material composition, repair instructions, and recycling guidance.

The core philosophy is to build a system that delivers transparency to consumers while remaining highly practical for businesses. Rather than exposing sensitive supply-chain data directly on a public ledger, the blockchain securely stores hashes, timestamps, and ownership records, whilst detailed documents are kept off-chain. This hybrid approach keeps the system cost-effective and realistic for companies guarding commercial secrets. The project aligns closely with the EU's Ecodesign for Sustainable Products Regulation (ESPR) and the broader European digital identity framework.

2 What problem(s) does it aim to solve?

EcoPassEU tackles several interconnected issues within sustainable consumption. Primarily, it combats 'greenwashing' by providing cryptographically verifiable evidence for environmental claims, allowing consumers to distinguish genuine sustainability from clever marketing.

Furthermore, as DPP requirements become mandatory across Europe, small and medium-sized enterprises (SMEs) face mounting regulatory pressure. EcoPassEU offers these businesses a low-cost, accessible tool to manage product data and demonstrate compliance without the need to build bespoke platforms. It also resolves data fragmentation across the supply chain by creating a single, tamper-evident source of truth. Ultimately, the DApp fosters trust among regulators needing auditable data, consumers seeking transparent proof, and firms requiring compliance tools that protect commercial confidentiality.

3 Who is the target audience?

The primary users are European manufacturers and brand owners—particularly SMEs in strictly regulated sectors—who require a practical solution to issue and update product compliance records.

Additionally, the DApp serves environmentally conscious consumers who want quick, reliable, and easily digestible summaries of a product’s lifecycle. Market surveillance bodies and regulators form another key audience, benefiting from structured, auditable records for spot checks and investigations. In the future, the platform’s utility could naturally extend to downstream stakeholders such as repair services and recyclers.

4 What techniques do you plan to use to implement the project?

At the blockchain layer, the backend will be built using Solidity smart contracts deployed on an EVM-compatible Proof-of-Stake network, such as Polygon, to ensure low gas costs and facilitate rapid iterative testing.

To represent product batches efficiently, we intend to implement the ERC-1155 semi-fungible token standard, which is better suited for batch-level tracking than minting individual NFTs for every physical item. For decentralised data storage, IPFS will host detailed documents (e.g., supplier certificates), with only the resulting content hashes committed to the blockchain. We also plan to integrate Decentralised Identifiers (DIDs) and Verifiable Credentials (VCs) to enable structured, authenticated sustainability claims, aligning with W3C standards. The frontend will be developed in React and connected via Ethers.js, prioritising a seamless user experience that abstracts Web3 complexity away from standard corporate users.

5 What is the key differentiating factor?

Unlike generic traceability prototypes that mandate radical transparency or lack practical compliance mechanisms, EcoPassEU is fundamentally grounded in privacy-aware regulatory alignment.

We recognise that enforcing absolute public disclosure of supply chain data is commercially unviable. By strictly separating on-chain cryptographic proofs from off-chain sensitive documents, we deliver verifiable sustainability without compromising corporate privacy. Furthermore, our project is deliberately scoped to be a realistic, targeted prototype rather than an overly ambitious, Europe-wide infrastructure, making it highly achievable within the constraints of this academic module.

6 What challenges or problems do you foresee?

Securing the physical-to-digital bridge remains a core challenge; while blockchain secures the digital record, physical QR codes or NFC tags can still be cloned or attached to counterfeit products.

Another critical issue is the inherent 'garbage in, garbage out' dilemma. Blockchain preserves records immutably but cannot verify the real-world truthfulness of the initial data entry, highlighting the need for robust off-chain governance. Additionally, we must carefully navigate the friction between GDPR and blockchain immutability, ensuring no sensitive data is written to the permanent ledger. Finally, achieving user adoption requires overcoming the technical intimidation associated with Web3 interfaces, necessitating a highly intuitive frontend design.

7 What questions do you have for the teaching staff?

- Is a simulated DPP workflow sufficient for this module, or is direct integration with existing external infrastructure expected?
- Is deploying to a testnet like Polygon Amoy (or a local Hardhat environment) acceptable for the prototype stage?
- How should we balance our focus between technical implementation and regulatory/conceptual alignment for grading purposes?
- Is demonstrating the physical link via a simulated QR code adequate, or should we incorporate physical NFC hardware?
- Does our proposed architecture (IPFS for documents, hashes on-chain) sufficiently address privacy and system design expectations for the final deliverable?

References

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